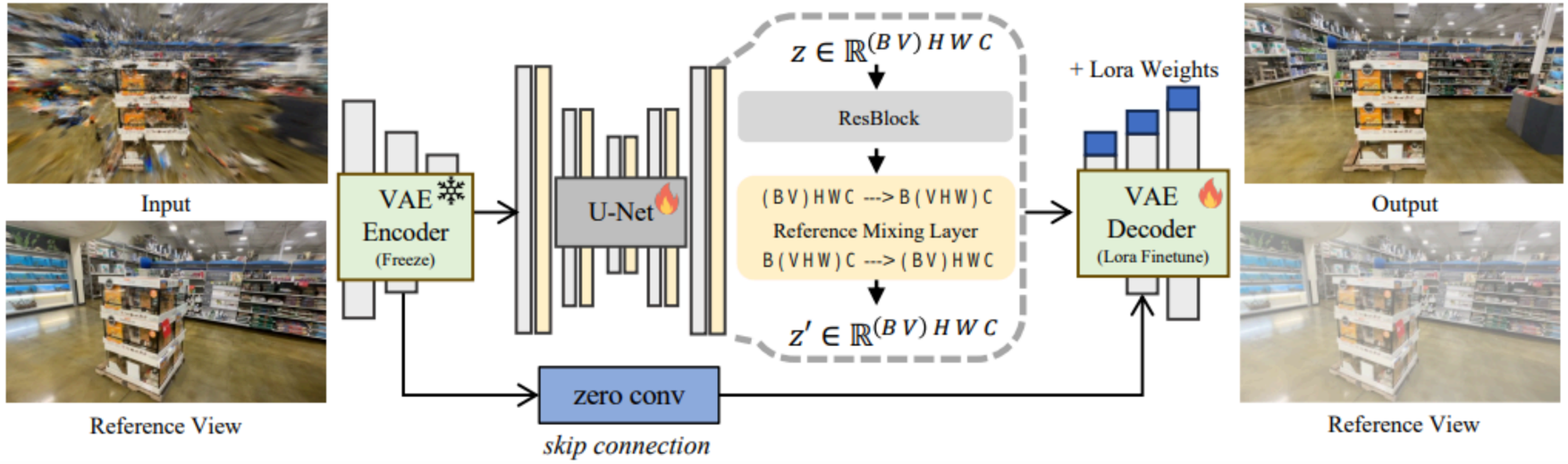
# DIFIX3D+: Improving 3D Reconstructions with Single-Step Diffusion Models

CVPR 2025

## Model Architecture




- NeRF / 3DGS first render a novel view from the reconstructed 3D scene, which may contain artifacts
- Single-step diffusion model for degraded to clean translation.
- Reference Mixing Layer ensures cross-view consistency.

## Algorithm

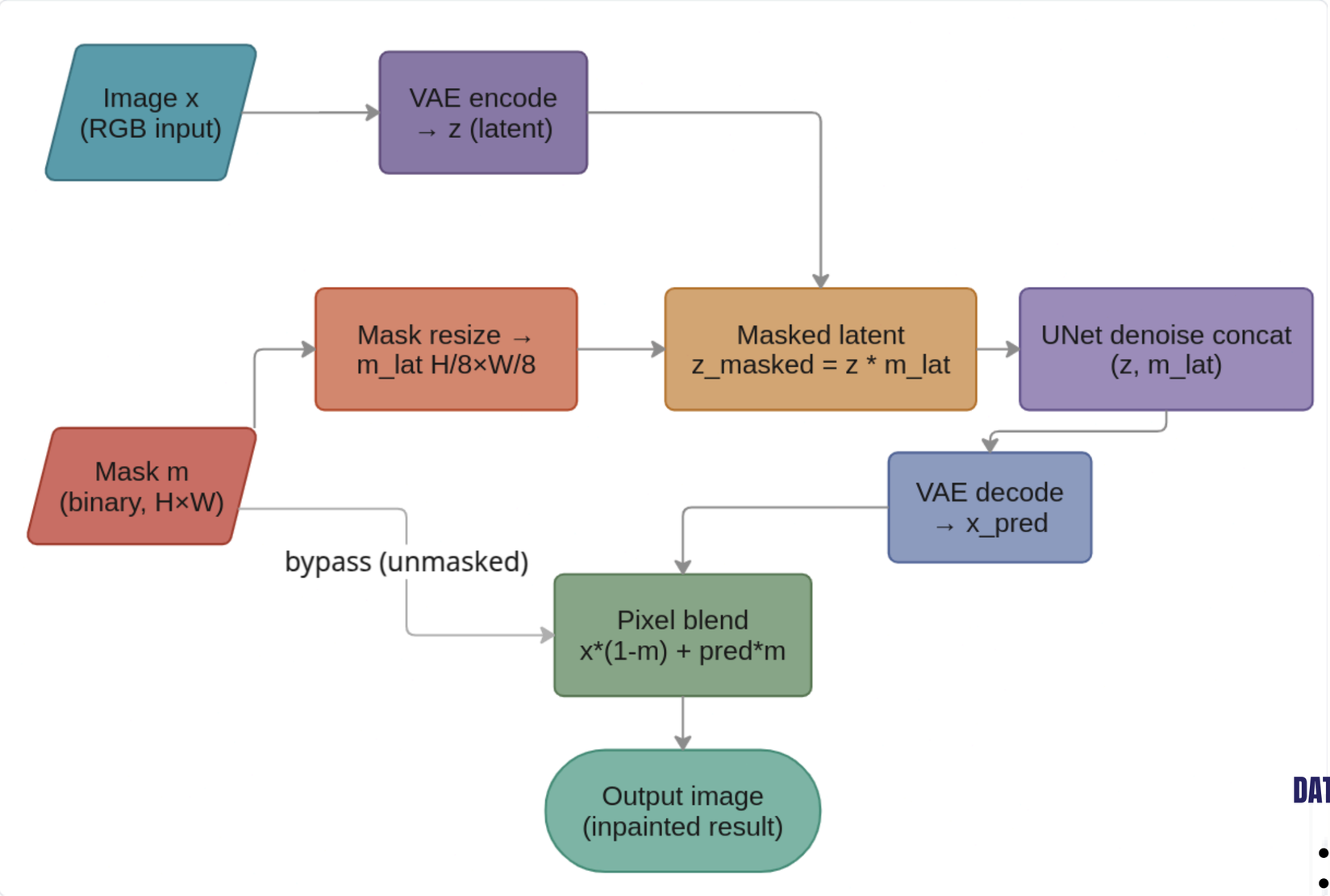
**Algorithm 1:** Progressive 3D Updates for Novel View Rendering

**Input:** Reference views $V_{\text{ref}}$, Target views $V_{\text{target}}$, 3D representation $R$ (e.g., NeRF, 3DGS), Diffusion model $D$ (DIFIX), Number of iterations per refinement $N_{\text{iter}}$, Perturbation step size $\Delta_{\text{pose}}$

**Output:** High-quality, artifact-free renderings at $V_{\text{target}}$

**Initialize:** Optimize 3D representation $R$ using $V_{\text{ref}}$.
**while** *not converged* **do**
  /* Optimize the 3D representation */
  **for** $i = 1$ *to* $N_{iter}$ **do**
    Optimize $R$ using the current training set.
  /* Generate novel views by perturbing camera poses */
  **for** ***each*** $v \in V_{target}$ **do**
    Find the nearest camera pose of $v$ in the training set.
    Perturb the nearest camera pose by $\Delta_{\text{pose}}$.
    Render novel view $\hat{v}$ using $R$.
    Refine $\hat{v}$ using diffusion model $D$.
    Add refined view $\hat{v}$ to the training set.
**return** Refined renderings at $V_{\text{target}}$.

# MASK PIPELINE




# TEXT GUIDANCE

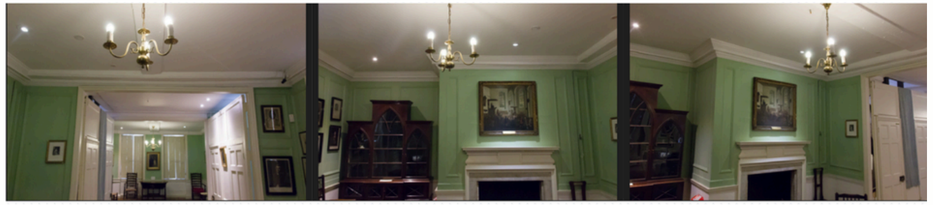
- using frozen decoders
- using pretrained 2d diffusion models for text based guidance after the vae decoder of difix3d

## DATASETS USED FOR OUR EXPERIMENTATIONS

- DL3DV
- NERSEMBLE 3D Real Face Dataset (TUM)
- Synthetic 3D Face Dataset (Microsoft Research)

# OUR EXPERIMENTS

## 3DGS



Reference Image a | Reference Image b | Reference Image c

3DGS output | Difix Output

Quantitative evaluation of Difix

| Metric | Value |
|---|---|
| PSNR | 10.386 |
| SSIM | 0.3381 |
| LPIPS | 0.6977 |
| FID | 43.47 |

## G3SPLAT

Reference Image a | Reference Image b | g3splat output | Difix output

## INPAINT RESULTS

Input Image | Reference Image | Binary Mask | Mask Overlayed

G3SPLAT output | DIFIX output no prompt given | DIFIX output prompt: make a smiling face

# CONTRIBUTORS

- SIRIN CHANGULANI @B23CM1038

Worked on mask-based inpainting, text-guided generation, and contributed to the project report.

- SONAM SIKARWAR @ B23CM1060

Performed experiments and inference with g3splat 3D model , preprocessed NerSemble , built demos, and created the final video.

- HARSHITA VACHHANI @B23EE1026

Evaluated Difix3D using quantitative metrics (PSNR, SSIM, LPIPS, FID) on the NeRF-Synthetic dataset, worked with 3DGS, and contributed to the report.